

# Lab 5

Emilien Zhang

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## Problem 1

### (a) Sampling the Sine Wave

$$x(t) = \sin(\Omega_0 t), \quad \Omega_0 = 2\pi(1000) \text{ rad/s}$$
$$\Omega_s = 2\pi(8192) \text{ rad/s}, \quad T = \frac{1}{8192} \text{ s}$$

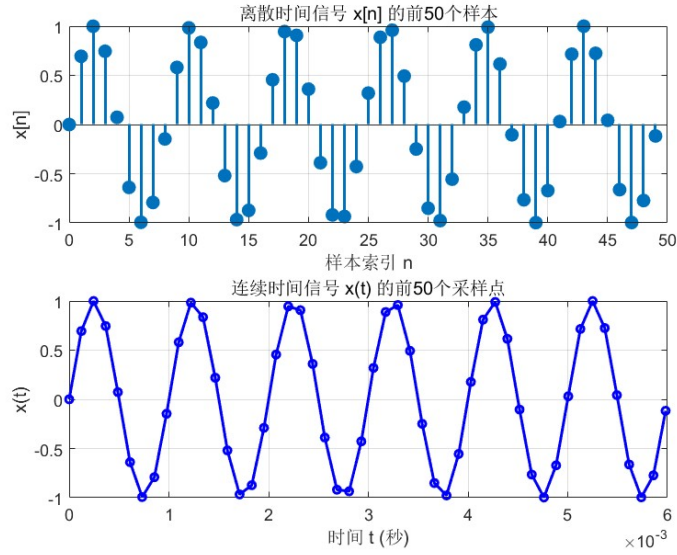
Since  $f_0 = 1000 \text{ Hz} < f_s/2 = 4096 \text{ Hz}$ , the Nyquist sampling theorem is satisfied and no aliasing occurs.

### (b) Displaying First 50 Samples

Two visualization methods are used:

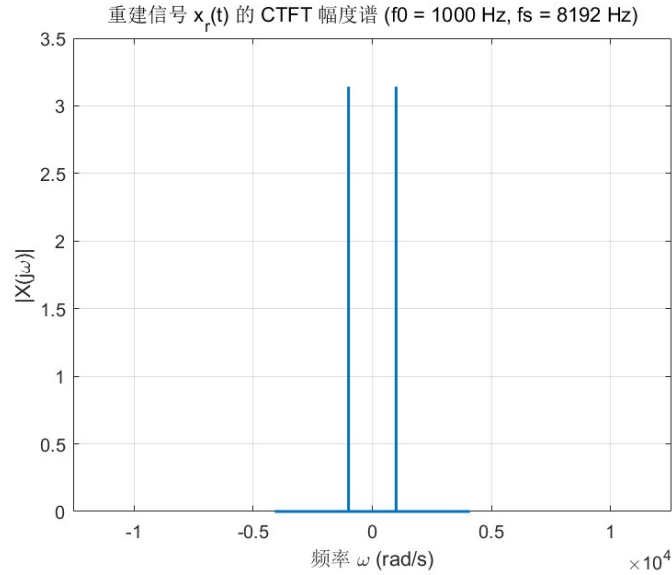
- **Stem plot:** displays the discrete sequence  $x[n]$  versus the sample index  $n$ . This emphasizes the discrete nature of the sampled signal.
- **Line plot:** connects the samples  $x(t)$  versus time  $t$  using straight lines. This approximates the continuous-time signal.

Both plots clearly show the sinusoidal waveform. Since the sampling rate is much higher than the signal frequency, the line plot closely matches the original continuous sine wave.



### (c) CTFT of the Reconstructed Signal

- **Magnitude spectrum:** exhibits two peaks at  $\Omega = \pm 2\pi(1000)$  rad/s. Due to numerical computation, these appear as narrow pulses rather than ideal impulses. All other values are negligibly small (round-off errors).



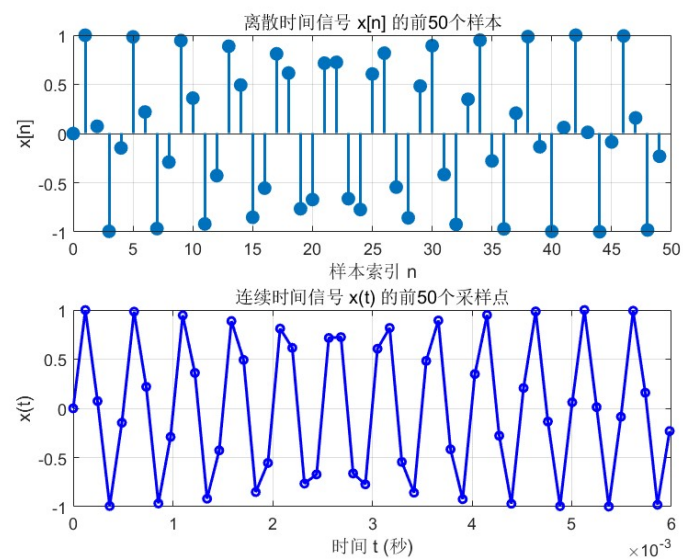
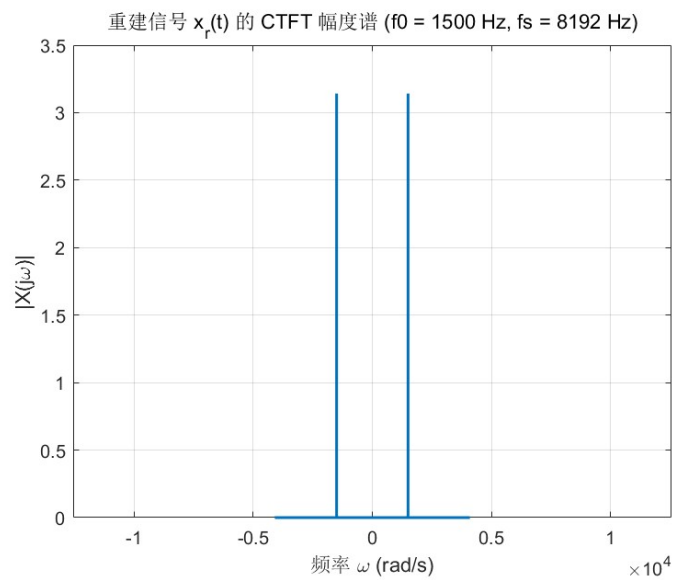
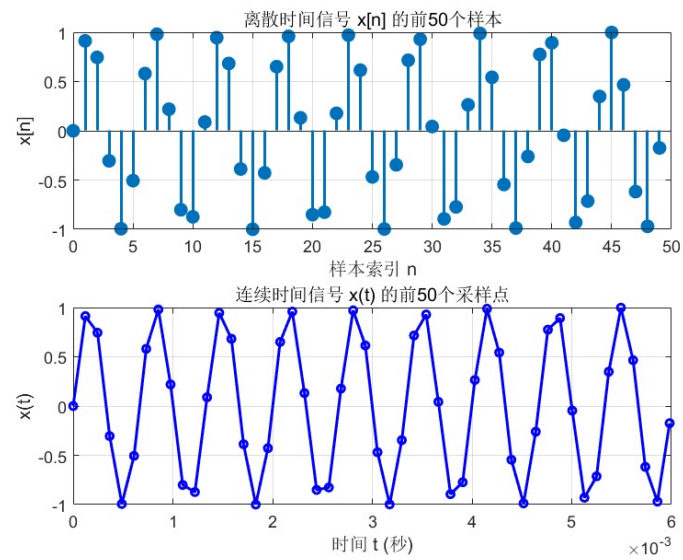
### (d) Frequencies 1500 Hz and 2000 Hz

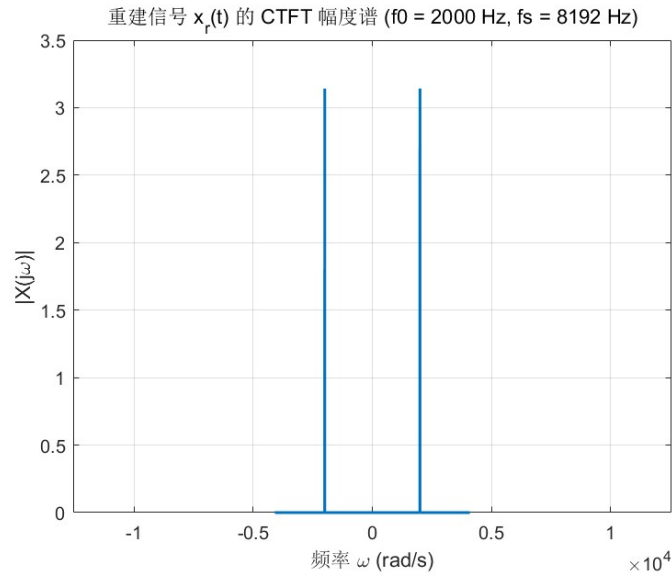
For  $\Omega_0 = 2\pi(1500)$  and  $\Omega_0 = 2\pi(2000)$  rad/s:

Both frequencies satisfy  $f_0 < 4096$  Hz, so no aliasing occurs.

- **Magnitude:** peaks appear at the correct frequencies ( $\pm 1500$  Hz and  $\pm 2000$  Hz)

The results confirm that signals below the Nyquist frequency are perfectly reconstructed.





### (e) Audio Pitch Test

When the sampled signals are played using the sound function at  $f_s = 8192$  Hz:

- The 1500 Hz signal produces a tone at 1500 Hz
- The 2000 Hz signal produces a tone at 2000 Hz
- The pitch increases as  $\Omega_0$  increases

This is expected because both frequencies are below the Nyquist frequency, so the reconstructed signal retains the original frequency.

### (f) High Frequencies and Aliasing

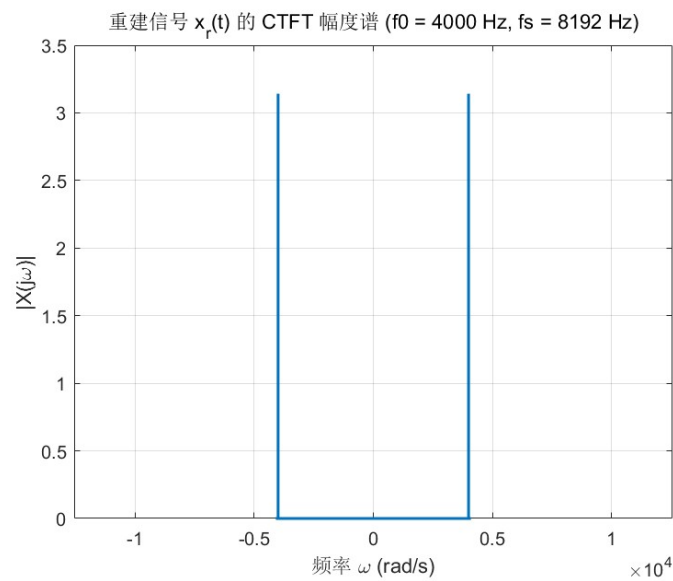
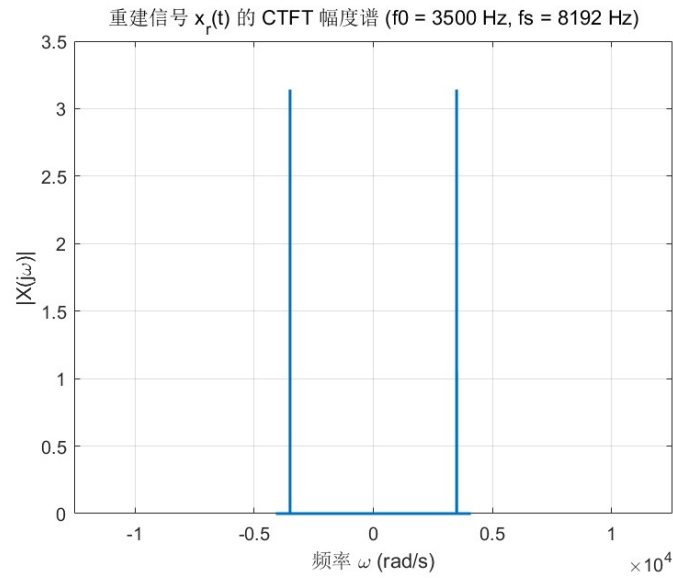
Now consider  $f_0 = 3500, 4000, 4500, 5000, 5500$  Hz.

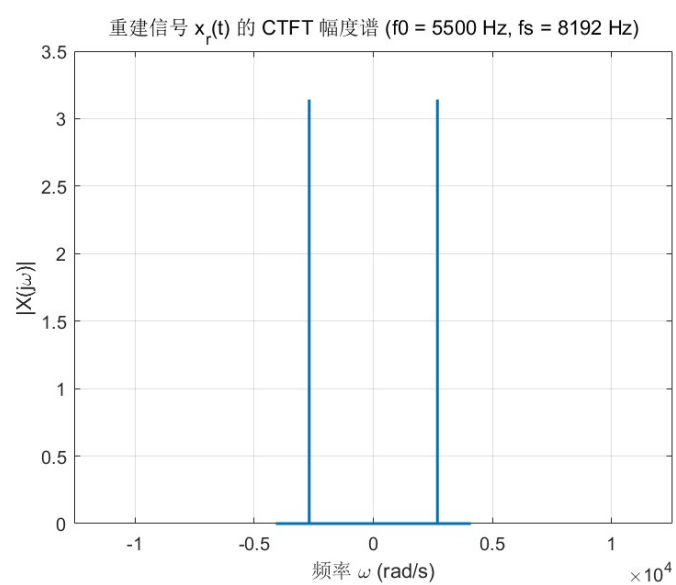
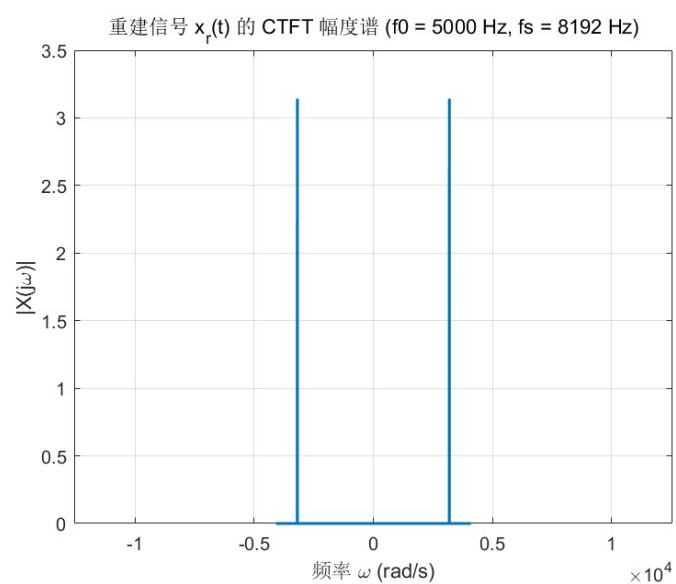
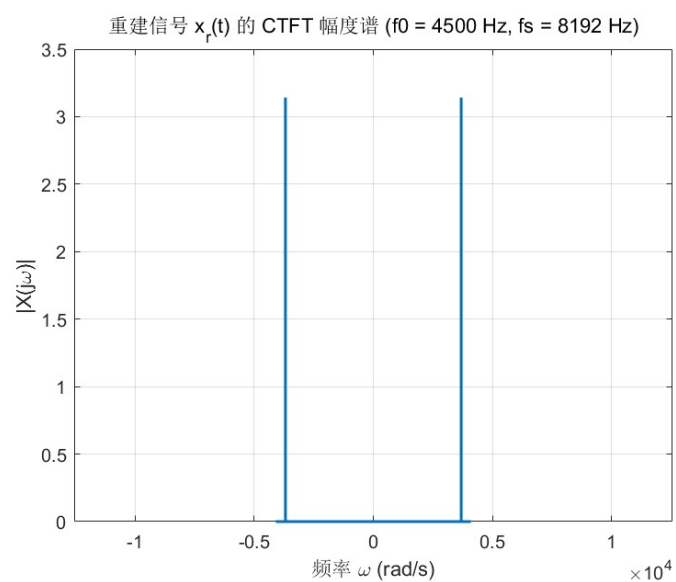
The Nyquist frequency is  $f_s/2 = 4096$  Hz. When  $f_0 > 4096$  Hz, aliasing occurs and the heard frequency is:

$$f_{\text{heard}} = |f_0 - kf_s|, \quad \text{where } f_{\text{heard}} \leq f_s/2$$

| $f_0$ (Hz) | Aliasing? | $f_{\text{heard}}$ (Hz) | Pitch Change |
|------------|-----------|-------------------------|--------------|
| 3500       | No        | 3500                    | ↑            |
| 4000       | No        | 4000                    | ↑            |
| 4500       | Yes       | $ 4500 - 8192  = 3692$  | ↓            |
| 5000       | Yes       | $ 5000 - 8192  = 3192$  | ↓            |
| 5500       | Yes       | $ 5500 - 8192  = 2692$  | ↓            |

**Key observation:** As  $f_0$  increases from 4500 Hz to 5500 Hz, the heard pitch decreases from 3692 Hz to 2692 Hz. This is the classic aliasing effect: high frequencies "fold back" into the low-frequency region.





## (g) Chirp Signal Sampling

The chirp signal is defined as:

$$x(t) = \sin\left(\Omega_0 t + \frac{1}{2}\beta t^2\right)$$

with parameters:

$$\Omega_0 = 2\pi(3000) \text{ rad/s}, \quad \beta = 2000 \text{ rad/s}^2$$

The instantaneous frequency is:

$$\Omega_{\text{inst}}(t) = \frac{d}{dt}\left(\Omega_0 t + \frac{1}{2}\beta t^2\right) = \Omega_0 + \beta t$$

In Hz:

$$f_{\text{inst}}(t) = \frac{\Omega_{\text{inst}}(t)}{2\pi} = 3000 + \frac{2000}{2\pi}t \text{ Hz}$$

For  $0 \leq t < 1$  s:

$$3000 \leq f_{\text{inst}}(t) \leq 3000 + \frac{2000}{2\pi} \approx 3318 \text{ Hz}$$

Since the instantaneous frequency remains below 4096 Hz throughout the interval, **no aliasing occurs**.

The sampled discrete signal is:

$$x[n] = \sin\left(2\pi \cdot \frac{3000}{8192} \cdot n + \frac{1000}{8192^2} \cdot n^2\right), \quad n = 0, 1, \dots, 8191$$

## (h) Heard Chirp Sound

When played, the chirp signal produces:

- A tone that starts at approximately 3000 Hz
- Smoothly rises to approximately 3318 Hz over 1 second
- Sounds like a short upward sweep or a bird chirp

No abrupt changes occur because the instantaneous frequency remains below the Nyquist frequency throughout the duration.

## (i) Predicting Maximum Pitch Time

For longer durations, the instantaneous frequency will eventually exceed the Nyquist frequency.

The maximum audible pitch occurs at the boundary where:

$$f_{\text{inst}}(t) = \frac{f_s}{2} = 4096 \text{ Hz}$$

Solving:

$$3000 + \frac{2000}{2\pi}t = 4096$$
$$t_{\max} = \frac{2\pi(4096 - 3000)}{2000} = \frac{2\pi \cdot 1096}{2000} \approx 3.44 \text{ s}$$

At this time, the instantaneous frequency equals the Nyquist frequency, and the heard pitch reaches its maximum (4096 Hz). Beyond this point, aliasing causes the perceived pitch to decrease.

## (j) 10-Second Chirp Analysis

For  $0 \leq t < 10$  s:

$$3000 \leq f_{\text{inst}}(t) \leq 3000 + \frac{2000}{2\pi} \cdot 10 \approx 6183 \text{ Hz}$$

The signal exhibits two distinct phases:

- **Phase 1** ( $0 \leq t < 3.44$  s):  $f_{\text{inst}} < 4096$  Hz, no aliasing. Pitch rises from 3000 Hz to 4096 Hz.
- **Phase 2** ( $3.44 \leq t < 10$  s):  $f_{\text{inst}} > 4096$  Hz, aliasing occurs. The heard frequency is:

$$f_{\text{heard}}(t) = 8192 - f_{\text{inst}}(t) = 8192 - \left(3000 + \frac{2000}{2\pi}t\right)$$

At  $t = 10$  s:

$$f_{\text{heard}}(10) = 8192 - 6183 \approx 2009 \text{ Hz}$$

Thus, the overall sound is:

- Pitch rises from 3000 Hz to 4096 Hz (first 3.44 s)
- Then pitch falls from 4096 Hz to about 2009 Hz (remaining time)

The zero-frequency (silence) point is predicted when:

$$f_{\text{inst}}(t) = f_s = 8192 \text{ Hz}$$
$$t_{\text{zero}} = \frac{2\pi(8192 - 3000)}{2000} \approx 16.31 \text{ s}$$

## Problem 2

This problem explores sinusoidal amplitude modulation (AM) and demodulation of speech signals, including synchronization between transmitter and receiver.



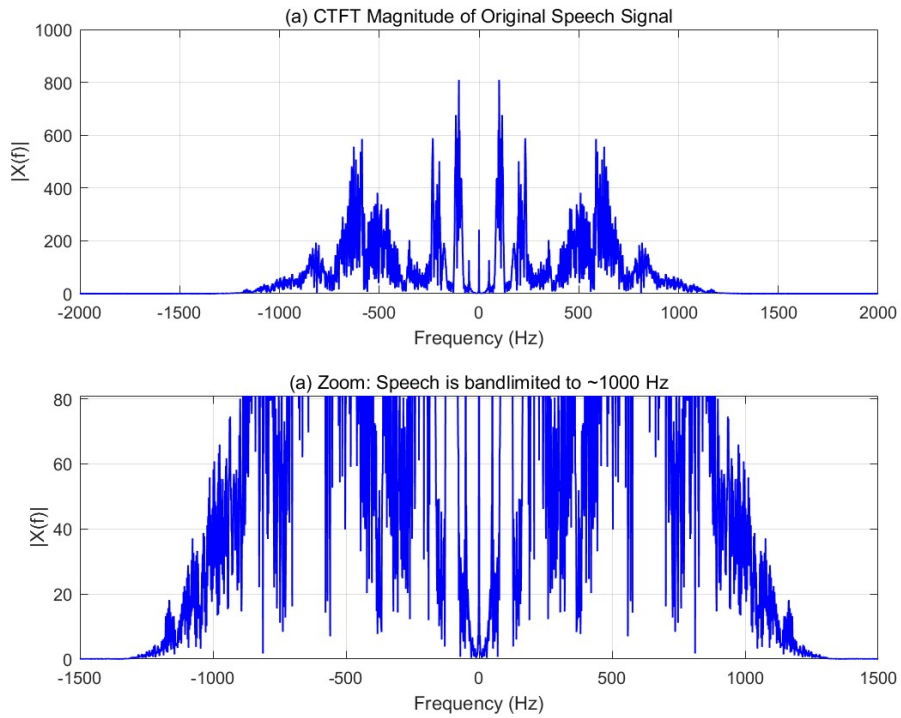
## (a) Speech Signal Spectrum

The speech signal  $x(t)$  is loaded from `origbl.mat`, sampled at  $f_s = 8192$  Hz. Its CTFT is computed via FFT:

$$X(f) = \mathcal{F}\{x(t)\}$$

The speech is bandlimited to approximately 1000 Hz:

$$X(f) \approx 0, \quad |f| > 1000 \text{ Hz}$$



## (b) AM Modulation

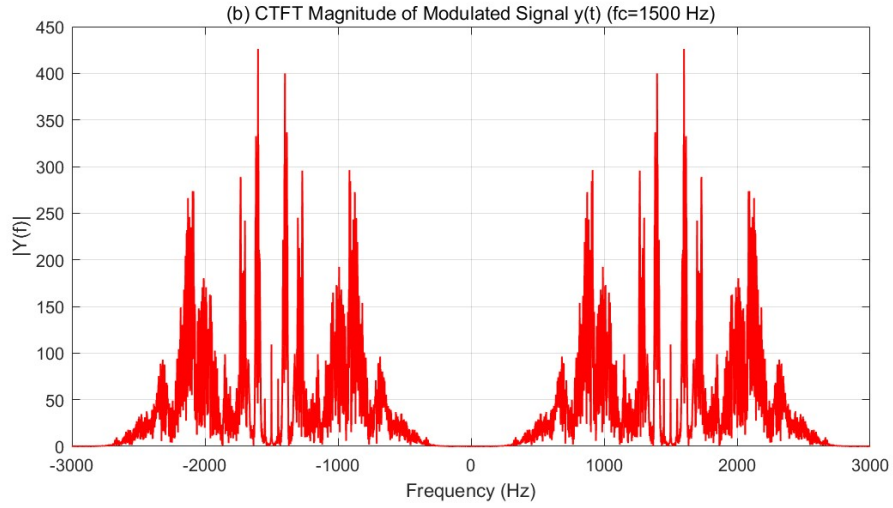
The modulated signal is:

$$y(t) = x(t) \cos(\omega_c t), \quad \omega_c = 2\pi f_c, \quad f_c = 1500 \text{ Hz}$$

Its spectrum is:

$$Y(f) = \frac{1}{2} [X(f - f_c) + X(f + f_c)]$$

The spectrum is shifted to  $\pm 1500$  Hz.



### (c) Synchronous Demodulation

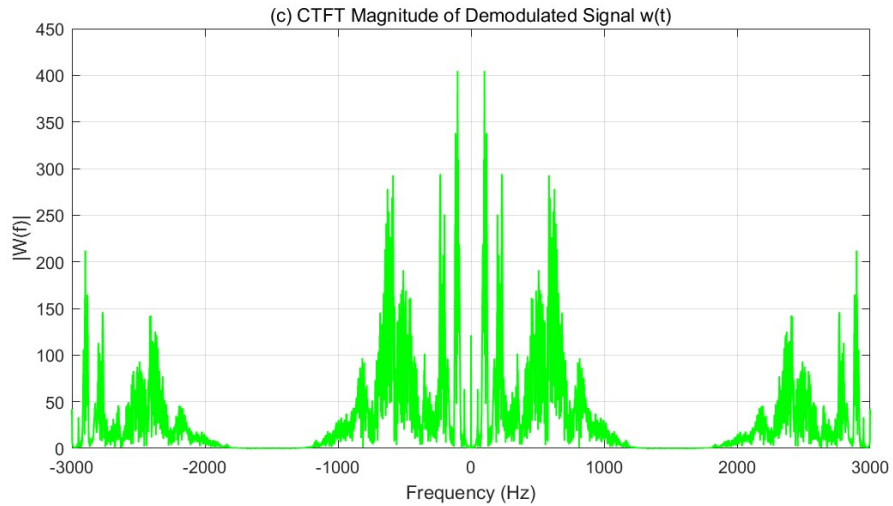
With perfect synchronization:

$$w(t) = y(t) \cos(\omega_c t) = x(t) \cos^2(\omega_c t) = \frac{1}{2}x(t) + \frac{1}{2}x(t) \cos(2\omega_c t)$$

Its spectrum:

$$W(f) = \frac{1}{2}X(f) + \frac{1}{4}[X(f - 2f_c) + X(f + 2f_c)]$$

Lowpass filtering removes the high-frequency components at  $\pm 2f_c$ , recovering  $\frac{1}{2}x(t)$ .



### (d) Lowpass Filter Design

The ideal lowpass filter has cutoff frequency  $f_c = 1500$  Hz:

$$H(f) = \begin{cases} 1, & |f| \leq f_c \\ 0, & |f| > f_c \end{cases}$$

Impulse response:

$$h(t) = 2f_c \cdot \text{sinc}(2f_c t) = \frac{\sin(2\pi f_c t)}{\pi t}$$

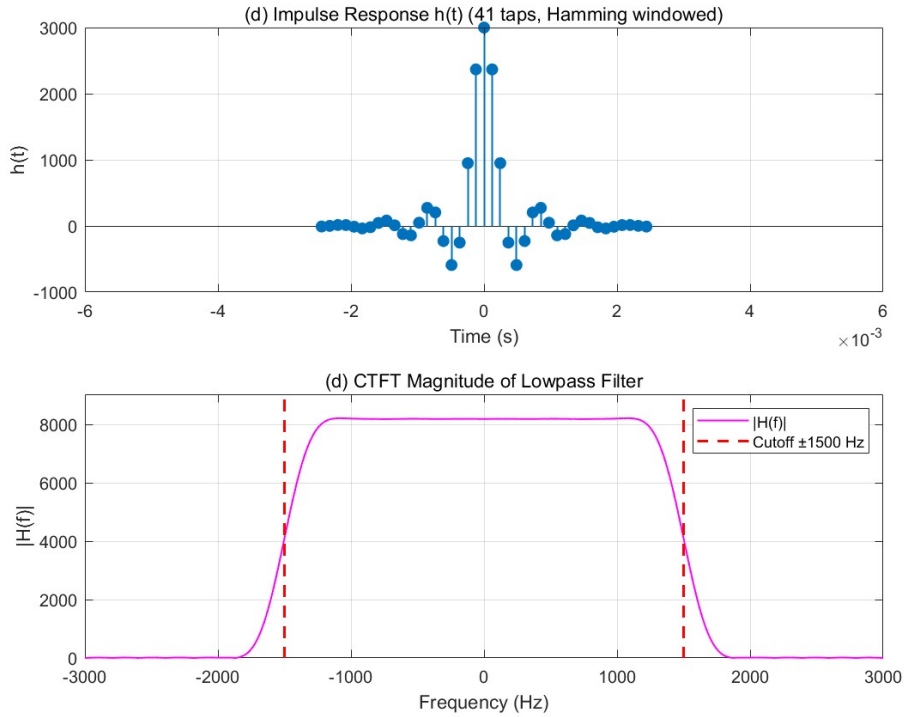
For discrete-time implementation with  $L = 41$  taps and sampling period  $T_s = 1/f_s$ :

$$h[n] = h(nT_s - T), \quad n = 0, 1, \dots, 40$$

where  $T = 20/f_s$  is the time shift to make the filter causal.

A Hamming window  $w[n]$  is applied:

$$h_w[n] = h[n] \cdot w[n], \quad w[n] = 0.54 - 0.46 \cos\left(\frac{2\pi n}{L-1}\right)$$



## (e) Speech Recovery

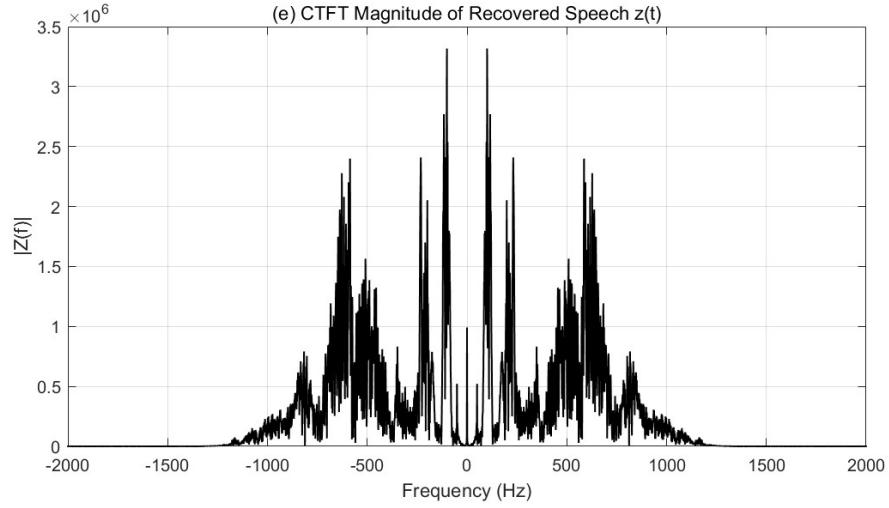
The demodulated signal is filtered:

$$z(t) = w(t) * h(t)$$

After filtering:

$$z(t) \approx \frac{1}{2}x(t)$$

The speech is recovered and played using `sound(z, fs)`.



## (f) Phase Mismatch Effects

With phase error  $\phi$ :

$$w_\phi(t) = y(t) \cos(\omega_c t + \phi) = x(t) \cos(\omega_c t) \cos(\omega_c t + \phi)$$

Using trigonometric identity:

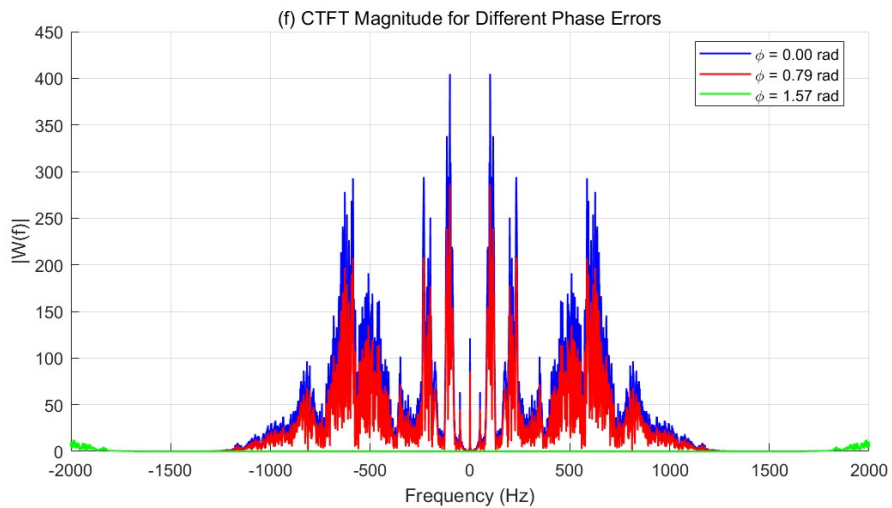
$$w_\phi(t) = \frac{x(t)}{2} [\cos(\phi) + \cos(2\omega_c t + \phi)]$$

After lowpass filtering:

$$z_\phi(t) = \frac{x(t)}{2} \cos(\phi)$$

| $\phi$  | $\cos(\phi)$               | Recovered Signal                       |
|---------|----------------------------|----------------------------------------|
| 0       | 1                          | $\frac{1}{2}x(t)$ (perfect)            |
| $\pi/4$ | $1/\sqrt{2} \approx 0.707$ | $\frac{1}{2\sqrt{2}}x(t)$ (attenuated) |
| $\pi/2$ | 0                          | 0 (no signal)                          |

When  $\phi = \pi/2$ , the signal is completely lost.



## (g) Phase-Locked Loop (PLL)

A DC offset  $A$  is added to ensure  $A + x(t) > 0$ :

$$y_{\text{PLL}}(t) = (A + x(t)) \cos(\omega_c t + \phi)$$

The PLL estimates the phase  $\hat{\phi}(t)$  using:

$$\hat{c}(k) = A \cos(\omega_c t_k + \hat{\phi}_k)$$

$$\hat{\phi}_{k+1} = \hat{\phi}_k + \alpha(y_{\text{PLL}}(t_k) - \hat{c}(k))$$

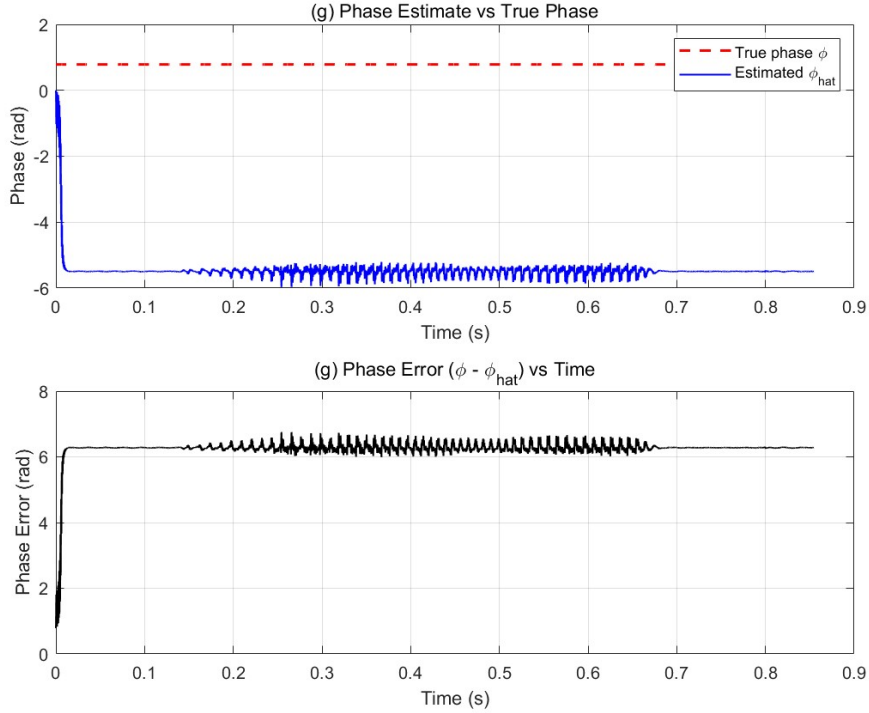
where  $\alpha$  is the adaptation step size.

The phase error converges to zero:

$$\lim_{k \rightarrow \infty} (\phi - \hat{\phi}_k) \rightarrow 0$$

Demodulation using the estimated phase:

$$w_{\text{PLL}}(t) = y_{\text{PLL}}(t) \cos(\omega_c t + \hat{\phi}(t))$$



## (h) Random Phase Jitter

The received phase includes random drift:

$$\phi_n = \sum_{i=1}^n \eta_i, \quad \eta_i \sim \mathcal{N}(0, \sigma^2)$$

where  $\sigma = 0.01$  is the RMS jitter.

Received signal:

$$y_{\text{jitter}}(t) = (A + x(t)) \cos(\omega_c t + \phi_n)$$

The PLL tracks the phase drift:

$$\hat{\phi}_{n+1} = \hat{\phi}_n + \alpha(y_{\text{jitter}}(t_n) - A \cos(\omega_c t_n + \hat{\phi}_n))$$

## Parameter Study

PLL performance depends on:

1. **Adaptation parameter  $\alpha$** : Larger  $\alpha \Rightarrow$  faster convergence but more noise.
2. **Carrier amplitude  $A$** : Larger  $A \Rightarrow$  stronger signal, easier lock.
3. **Phase jitter  $\sigma$** : Larger  $\sigma \Rightarrow$  harder to track.

